

ISC Class XII Physics

Chapter 6 Electromagnetic Induction

20 Numericals Answer Key

Step-by-Step ISC Board Solutions

Topics Covered

- I. Magnetic Flux and Faraday's Laws
- II. Motional EMF
- III. Self-Inductance
- IV. Mutual Induction

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Formula Sheet

I. Magnetic Flux and Faraday's Laws

Formula: $\Phi = BA \cos \theta$

Φ = magnetic flux, B = field strength, A = area, θ = angle between B and the normal to the coil.

Formula: $\varepsilon = -N \frac{d\Phi}{dt}$

ε = induced emf (Faraday's Law), N = number of turns, $d\Phi/dt$ = rate of change of flux per turn.

Formula: $q = \frac{N\Delta\Phi}{R}$

q = induced charge, $N\Delta\Phi$ = change in flux linkage, R = total circuit resistance.

Formula: $M = \mu_0 n N A$, $\varepsilon = M \frac{dI}{dt}$

M = mutual inductance of a small coil (N turns, area A) wound on a long solenoid with n turns/metre.

II. Motional EMF

Formula: $\varepsilon = Bvl$

ε = motional emf, v = velocity of the conductor, l = length of conductor, valid when v , B , l are mutually perpendicular.

Formula: $t = \frac{d}{v}$

Duration for which emf lasts while a loop of side d (measured along the direction of motion) leaves a field region at speed v .

III. Self-Inductance

Formula: $\varepsilon = -L \frac{dI}{dt}$

ε = self-induced emf, L = coefficient of self-induction, dI/dt = rate of change of current.

Formula: $L = \frac{N\Phi}{I}$

L = self-inductance, $N\Phi$ = total flux linkage of the coil, I = current producing it.

Formula:
$$M = \frac{\mu_0 N_1 N_2 A}{l}$$

Mutual inductance of two coils (N_1 , N_2 turns) wound on the same solenoid of length l and cross-section A .

IV. Mutual Induction

Formula:
$$M = \frac{\phi_1}{I_2}, \quad B_2 = \frac{\mu_0 I_2 R_2^2}{2(R_2^2 + x^2)^{3/2}}$$

M = mutual inductance between two coaxial circular loops; R_2 = radius of larger loop, x = distance between centres, found via reciprocity by placing current I_2 in the larger loop and computing flux through the smaller loop.

Note: $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$ (also written $\text{Wb A}^{-1} \text{ m}^{-1}$ or T m A^{-1} or N A^{-2}).

I. Magnetic Flux and Faraday's Laws

Question 1

A coil of wire enclosing an area 100 cm^2 is placed with its plane making an angle 60° with the magnetic field of strength 10^{-1} T . What is the flux through the coil? If magnetic field is reduced to zero in 10^{-3} s , then find the induced emf.

Given: $A = 100 \text{ cm}^2 = 10^{-2} \text{ m}^2$; angle between plane and field $= 60^\circ \Rightarrow$ angle with normal, $\theta = 30^\circ$; $B = 10^{-1} \text{ T}$; $t = 10^{-3} \text{ s}$

Formula: $\Phi = BA \cos \theta$; $\varepsilon = \frac{\Delta \Phi}{\Delta t}$

Solution:

$$\Phi = (10^{-1})(10^{-2}) \cos 30^\circ = 10^{-3} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2} \times 10^{-3} \text{ Wb}$$

When B is reduced to zero, the flux also falls from Φ to 0 in $t = 10^{-3} \text{ s}$:

$$\varepsilon = \frac{\Delta \Phi}{\Delta t} = \frac{\frac{\sqrt{3}}{2} \times 10^{-3} - 0}{10^{-3}} = \frac{\sqrt{3}}{2} = 0.866 \text{ V}$$

Final Answer

$$\Phi_m = \frac{\sqrt{3}}{2} \times 10^{-3} \text{ Wb}, \quad \text{induced emf} = 0.866 \text{ V}$$

Common Mistake: Always use the angle between B and the *normal* to the coil in $\cos \theta$, not the angle given between B and the plane of the coil – the two are complementary.

Question 2

The magnetic flux through a coil of 50 turns changes from 0.3 Wb to zero in 1 second. Calculate the induced emf across the two ends of the coil.

Given: $N = 50$; $\Delta \Phi = 0.3 - 0 = 0.3 \text{ Wb}$; $\Delta t = 1 \text{ s}$

Formula: $\varepsilon = N \frac{\Delta \Phi}{\Delta t}$

Solution:

$$\varepsilon = 50 \times \frac{0.3}{1} = 15 \text{ V}$$

Final Answer

$$\varepsilon = 15 \text{ V}$$

Question 3

A coil of area 500 cm^2 and 2000 turns is kept normal to a magnetic field of $4 \times 10^{-5} \text{ Wb m}^{-2}$. It is rotated through 180° in 0.1 s. (i) Calculate the average emf induced. (ii) If the total resistance of the circuit be 20Ω , calculate the induced charge.

Given: $A = 500 \text{ cm}^2 = 0.05 \text{ m}^2$; $N = 2000$; $B = 4 \times 10^{-5} \text{ Wb m}^{-2}$; $t = 0.1 \text{ s}$;
 $R = 20 \Omega$

Formula: $\varepsilon = \frac{2NBA}{t}$; $q = \frac{N\Delta\Phi_{\text{per turn}}}{R}$

Solution:

On rotating through 180° , the flux linkage reverses direction, so the total change in flux linkage is $2NBA$ (from $+NBA$ to $-NBA$):

$$NBA = 2000 \times (4 \times 10^{-5}) \times 0.05 = 4 \times 10^{-3} \text{ Wb-turns}$$

$$\varepsilon = \frac{2NBA}{t} = \frac{2 \times 4 \times 10^{-3}}{0.1} = 0.08 \text{ V}$$

For charge, the flux (per turn) reverses by $2BA$:

$$2BA = 2 \times (4 \times 10^{-5}) \times 0.05 = 4 \times 10^{-6} \text{ Wb}$$

$$q = \frac{N(2BA)}{R} = \frac{2000 \times 4 \times 10^{-6}}{20} = 4 \times 10^{-4} \text{ C}$$

Final Answer

(i) $\varepsilon = 0.08 \text{ V}$; (ii) $q = 4 \times 10^{-4} \text{ C}$

Common Mistake: On a 180° rotation the flux does not go to zero – it reverses sign. Forgetting the factor of 2 (i.e. using NBA instead of $2NBA$) is a very common ISC-board error.

Question 4

A coil of area 0.5 m^2 and having 100 turns is placed perpendicular to a magnetic field of 0.04 Wb m^{-2} . If the coil is suddenly removed away from the field in 0.01 second, calculate the average emf induced in the coil.

Given: $A = 0.5 \text{ m}^2$; $N = 100$; $B = 0.04 \text{ Wb m}^{-2}$; $t = 0.01 \text{ s}$

Formula: $\varepsilon = \frac{NBA}{t}$

Solution:

$$\varepsilon = \frac{100 \times 0.04 \times 0.5}{0.01} = \frac{2}{0.01} = 200 \text{ V}$$

Final Answer

$$\varepsilon = 200 \text{ V}$$

Question 5

A 28 turn coil with average diameter 0.02 m is placed perpendicular to a magnetic field of 8000 T. If the field changes to 3000 T in 4 s, what is the magnitude of the induced emf? ($\pi = 22/7$)

Given: $N = 28$; diameter = 0.02 m $\Rightarrow r = 0.01$ m; $B_1 = 8000$ T, $B_2 = 3000$ T;
 $t = 4$ s

Formula: $\varepsilon = NA \frac{\Delta B}{\Delta t}$

Solution:

$$A = \pi r^2 = \frac{22}{7} \times (0.01)^2 = 3.143 \times 10^{-4} \text{ m}^2$$

$$\Delta B = 8000 - 3000 = 5000 \text{ T}$$

$$\varepsilon = 28 \times (3.143 \times 10^{-4}) \times \frac{5000}{4} = \frac{28 \times 3.143 \times 10^{-4} \times 5000}{4} = 11 \text{ V}$$

Final Answer

$$\varepsilon = 11 \text{ V}$$

Question 6

A closed copper coil of area 1.0 m² placed perpendicular to a magnetic field of 0.10 T is rotated through 180° in 0.01 s. Find the induced emf and induced current in the coil whose resistance is 2.0 Ω . If the coil is open somewhere, then what will be the emf and the current?

Given: $A = 1.0 \text{ m}^2$; $B = 0.10 \text{ T}$; $t = 0.01 \text{ s}$; $R = 2.0 \Omega$

Formula: $\varepsilon = \frac{2BA}{t}$; $I = \frac{\varepsilon}{R}$

Solution:

Rotation by 180° reverses the flux, so the total change is $2BA$:

$$\varepsilon = \frac{2 \times 0.10 \times 1.0}{0.01} = \frac{0.2}{0.01} = 20 \text{ V}$$

$$I = \frac{\varepsilon}{R} = \frac{20}{2.0} = 10 \text{ A}$$

If the coil is open (infinite resistance), the induced emf will still be 20 V, because emf depends only on the rate of change of flux and not on the resistance of the circuit; however the current will be zero since no closed path exists for charge to flow.

Final Answer

Closed coil: $\varepsilon = 20 \text{ V}$, $I = 10 \text{ A}$. Open coil: $\varepsilon = 20 \text{ V}$, $I = 0$

Common Mistake: Students often assume emf becomes zero when the circuit is broken – emf is a property of the changing flux; only current requires a closed path.

Question 7

A rectangular loop of sides $10 \text{ cm} \times 3 \text{ cm}$ with a small cut is moving out of a uniform magnetic field of 0.5 T directed normal to the loop. Compute the voltage induced across the cut, if the velocity of the loop is 1 cm s^{-1} in a direction (i) normal to the longer side, (ii) normal to the shorter side of the loop. For how long does the induced voltage last in each case?

★ Likely Exam Question

Given: $l_1 = 10 \text{ cm} = 0.1 \text{ m}$ (longer side); $l_2 = 3 \text{ cm} = 0.03 \text{ m}$ (shorter side);
 $B = 0.5 \text{ T}$; $v = 1 \text{ cm s}^{-1} = 0.01 \text{ m s}^{-1}$

Formula: $\varepsilon = Bvl$; $t = \frac{d}{v}$

Solution:

(i) Velocity normal to the longer side: here it is the longer side (length 0.1 m) that sweeps across the field boundary, so this is the effective length in $\varepsilon = Bvl$. The loop must travel a distance equal to the shorter side (0.03 m) to completely exit the field.

$$\varepsilon_1 = Bvl_1 = 0.5 \times 0.01 \times 0.1 = 5 \times 10^{-4} \text{ V} = 0.5 \text{ mV}$$

$$t_1 = \frac{l_2}{v} = \frac{0.03}{0.01} = 3 \text{ s}$$

(ii) Velocity normal to the shorter side: now the shorter side (length 0.03 m) sweeps the boundary, and the loop must travel a distance equal to the longer side (0.1 m) to exit.

$$\varepsilon_2 = Bvl_2 = 0.5 \times 0.01 \times 0.03 = 1.5 \times 10^{-4} \text{ V} = 0.15 \text{ mV}$$

$$t_2 = \frac{l_1}{v} = \frac{0.1}{0.01} = 10 \text{ s}$$

Final Answer

(i) $\varepsilon_1 = 0.5 \text{ mV}$, lasts 3 s; (ii) $\varepsilon_2 = 0.15 \text{ mV}$, lasts 10 s

Common Mistake: The side that generates the emf (Bvl) and the side that decides the duration (d/v) are the two *different* sides of the rectangle – mixing them up is the most common slip in this question.

Question 8

A long solenoid of radius 0.01 m has 600 turns per metre. A smaller coil of 100 turns is wrapped closely around the centre of the bigger solenoid. If the current in the solenoid rises from zero to 2 A in 1 ms, find the emf induced in the smaller coil.

Given: $r = 0.01 \text{ m}$; $n = 600 \text{ turns/m}$; $N = 100$; $\Delta I = 2 \text{ A}$; $\Delta t = 1 \text{ ms} = 10^{-3} \text{ s}$

Formula: $M = \mu_0 n N A$; $\varepsilon = M \frac{dI}{dt}$

Solution:

$$A = \pi r^2 = \pi (0.01)^2 = 3.1416 \times 10^{-4} \text{ m}^2$$

$$M = \mu_0 n N A = (4\pi \times 10^{-7})(600)(100)(3.1416 \times 10^{-4}) = 2.369 \times 10^{-5} \text{ H}$$

$$\varepsilon = M \frac{\Delta I}{\Delta t} = (2.369 \times 10^{-5}) \times \frac{2}{10^{-3}} = 0.0473 \text{ V} = 47.3 \text{ mV}$$

Final Answer

$\varepsilon = 47.3 \text{ mV}$

Question 9

When a conducting loop of resistance 10Ω and area 10 cm^2 is removed from an external magnetic field acting normally, the variation of induced current in the loop with time is shown in the figure (current falls linearly from 0.4 A at $t = 0$ to 0 at $t = 1.0 \text{ s}$). Find (i) total charge passed through the loop, (ii) change in magnetic flux through the loop, (iii) magnitude of the magnetic field applied.

Given: $R = 10 \Omega$; $A = 10 \text{ cm}^2 = 10^{-3} \text{ m}^2$; I falls linearly from 0.4 A to 0 over 1.0 s

Formula: $q = \text{area under } I-t \text{ graph}$; $q = \frac{\Delta \Phi}{R}$; $B = \frac{\Delta \Phi}{A}$

Solution:

The charge equals the area of the triangle under the $I-t$ graph:

$$q = \frac{1}{2} \times 1.0 \times 0.4 = 0.2 \text{ C}$$

$$\Delta\Phi = qR = 0.2 \times 10 = 2 \text{ Wb}$$

$$B = \frac{\Delta\Phi}{A} = \frac{2}{10^{-3}} = 2000 \text{ T}$$

Final Answer

(i) $q = 0.2 \text{ C}$; (ii) $\Delta\Phi = 2 \text{ Wb}$; (iii) $B = 2000 \text{ T}$

Question 10

A 50 cm long horizontal wire is falling down with a speed of 6 m s^{-1} perpendicular to a uniform magnetic field of 1.1 Wb m^{-2} . The magnetic field is directed from east to west. Calculate the emf induced in the wire. Which end of the wire (north or south) will be positive?

Given: $l = 50 \text{ cm} = 0.5 \text{ m}$; $v = 6 \text{ m s}^{-1}$; $B = 1.1 \text{ Wb m}^{-2}$

Formula: $\varepsilon = Bvl$

Solution:

$$\varepsilon = Bvl = 1.1 \times 6 \times 0.5 = 3.3 \text{ V}$$

By the motional-emf (Lorentz force) rule, the force on free electrons in the falling wire drives negative charge towards the south end; hence the north end of the wire is left positive.

Final Answer

$\varepsilon = 3.3 \text{ V}$. Under the Lorentz force, negative charge accumulates at the south end, so the **north end** of the wire is positive.

Question 11

The axle of a car is 2.4 m long. If the car is going straight with a speed of 25 m s^{-1} towards south and the vertical component of earth's magnetic field at that place is $3 \times 10^{-5} \text{ T}$, then calculate the potential difference induced between the ends of the axle.

Given: $l = 2.4 \text{ m}$; $v = 25 \text{ m s}^{-1}$; $B_v = 3 \times 10^{-5} \text{ T}$

Formula: $\varepsilon = Bvl$

Solution:

$$\varepsilon = Bvl = (3 \times 10^{-5}) \times 25 \times 2.4 = 1.8 \times 10^{-3} \text{ V}$$

Final Answer

$\varepsilon = 1.8 \times 10^{-3} \text{ V}$

II. Motional EMF

Question 12

A 50 cm long wire is moving in a uniform magnetic field of $0.8 \text{ N A}^{-1} \text{ m}^{-1}$ with a velocity of 2 m s^{-1} perpendicular to both its length and the magnetic field. Calculate the induced emf in the wire. If the wire be moving parallel to the field, what will be the value of the induced emf?

Given: $l = 0.5 \text{ m}$; $B = 0.8 \text{ T}$; $v = 2 \text{ m s}^{-1}$

Formula: $\varepsilon = Bvl$

Solution:

When $v \perp B \perp l$:

$$\varepsilon = Bvl = 0.8 \times 2 \times 0.5 = 0.8 \text{ V}$$

If the wire moves parallel to the field, v and B are along the same line, so the component of v perpendicular to B is zero; hence no motional emf is induced.

Final Answer

$\varepsilon = 0.8 \text{ V}$ (perpendicular motion); $\varepsilon = 0$ (motion parallel to field)

Common Mistake: Motional emf depends on the component of velocity perpendicular to B ; motion *along* the field lines produces zero emf regardless of speed.

Question 13

A 0.4 m long straight conductor is moved in a magnetic field of induction 0.9 Wb m^{-2} with a velocity of 7 m s^{-1} . Calculate the maximum emf induced in the conductor.

Given: $l = 0.4 \text{ m}$; $B = 0.9 \text{ Wb m}^{-2}$; $v = 7 \text{ m s}^{-1}$

Formula: $\varepsilon_{\max} = Bvl$

Solution:

$$\varepsilon_{\max} = Bvl = 0.9 \times 7 \times 0.4 = 2.52 \text{ V}$$

Final Answer

$\varepsilon_{\max} = 2.52 \text{ V}$

III. Self-Inductance

Question 14

A plot of magnetic flux (ϕ) versus current (I) is shown in the figure for two inductors A and B , making angles of 60° and 30° respectively with the current axis. Find the ratio of the coefficient of self-inductance of A and B .

Given: Line A makes 60° with the I -axis; line B makes 30° with the I -axis

Formula: $L = \frac{\phi}{I} = \tan \theta$ (slope of the ϕ - I graph)

Solution:

The self-inductance is the slope of the ϕ versus I line:

$$L_A = \tan 60^\circ = \sqrt{3}, \quad L_B = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

$$\frac{L_A}{L_B} = \frac{\sqrt{3}}{1/\sqrt{3}} = 3$$

Final Answer

$$L_A : L_B = 3 : 1$$

Question 15

A magnetic flux of $5 \mu\text{Wb}$ is linked with a coil when a current of 1 mA flows through it. Calculate the self-inductance of the coil.

Given: $\phi = 5 \mu\text{Wb} = 5 \times 10^{-6} \text{ Wb}$; $I = 1 \text{ mA} = 10^{-3} \text{ A}$

Formula: $L = \frac{\phi}{I}$

Solution:

$$L = \frac{5 \times 10^{-6}}{10^{-3}} = 5 \times 10^{-3} \text{ H} = 5 \text{ mH}$$

Final Answer

$$L = 5 \text{ mH}$$

Question 16

If a rate of change of current of 2 A s^{-1} induces an emf of 10 mV in a solenoid, what is the self-inductance of the solenoid?

Given: $\frac{dI}{dt} = 2 \text{ A s}^{-1}; \quad \varepsilon = 10 \text{ mV} = 10 \times 10^{-3} \text{ V}$

Formula: $\varepsilon = L \frac{dI}{dt}$

Solution:

$$L = \frac{\varepsilon}{dI/dt} = \frac{10 \times 10^{-3}}{2} = 5 \times 10^{-3} \text{ H} = 5 \text{ mH}$$

Final Answer

$$L = 5 \text{ mH}$$

Question 17

The current in a coil increases from zero to 2 A in 0.05 s, due to which an induced emf of 80 V is developed. Calculate the coefficient of self-induction of the coil.

Given: $\Delta I = 2 \text{ A}; \quad \Delta t = 0.05 \text{ s}; \quad \varepsilon = 80 \text{ V}$

Formula: $\varepsilon = L \frac{\Delta I}{\Delta t}$

Solution:

$$L = \frac{\varepsilon \Delta t}{\Delta I} = \frac{80 \times 0.05}{2} = \frac{4}{2} = 2 \text{ H}$$

Final Answer

$$L = 2 \text{ H}$$

Question 18

A coil has an inductance of 4.0 H. Find the emf induced, if a current of 2.7 A in it is stopped within 1.35 s.

Given: $L = 4.0 \text{ H}; \quad \Delta I = 2.7 \text{ A}; \quad \Delta t = 1.35 \text{ s}$

Formula: $\varepsilon = L \frac{\Delta I}{\Delta t}$

Solution:

$$\varepsilon = 4.0 \times \frac{2.7}{1.35} = 4.0 \times 2 = 8.0 \text{ V}$$

Final Answer

$$\varepsilon = 8.0 \text{ V}$$

Question 19

A solenoid has 2000 turns wound over a length of 0.3 m. The area of cross-section is $1.2 \times 10^{-3} \text{ m}^2$. Around its central section a coil of 300 turns is closely wound. If an initial current of 2 A is reversed in 0.25 s, find the emf induced in the coil.

Given: $N_1 = 2000$; $l = 0.3 \text{ m}$; $A = 1.2 \times 10^{-3} \text{ m}^2$; $N_2 = 300$; current reversed from +2 A to -2 A in 0.25 s

Formula: $M = \frac{\mu_0 N_1 N_2 A}{l}$; $\varepsilon = M \frac{dI}{dt}$

Solution:

$$M = \frac{(4\pi \times 10^{-7})(2000)(300)(1.2 \times 10^{-3})}{0.3} = 3.016 \times 10^{-3} \text{ H}$$

Since the current is *reversed*, the net change is $\Delta I = 2 - (-2) = 4 \text{ A}$:

$$\frac{dI}{dt} = \frac{4}{0.25} = 16 \text{ A s}^{-1}$$

$$\varepsilon = M \frac{dI}{dt} = (3.016 \times 10^{-3}) \times 16 = 0.0483 \text{ V} = 48 \text{ mV}$$

Final Answer

$$\varepsilon = 48 \text{ mV}$$

Common Mistake: When a current is *reversed* (not just switched off), the change ΔI is the sum of the initial and final magnitudes ($I_1 + I_2$), not just I_1 – a very common ISC trap.

IV. Mutual Induction

Question 20

A circular loop of radius 0.3 cm lies parallel to a much bigger circular loop of radius 20 cm. The centre of the small loop is on the axis of the bigger loop. The distance between their centres is 15 cm. If a current of 2.0 A flows through the smaller loop, then calculate the flux linked with the bigger loop.

★ Likely Exam Question

Given: $r_1 = 0.3 \text{ cm} = 3 \times 10^{-3} \text{ m}$ (small loop); $R_2 = 20 \text{ cm} = 0.2 \text{ m}$ (big loop);
 $x = 15 \text{ cm} = 0.15 \text{ m}$; $I_1 = 2.0 \text{ A}$

Formula: $M = \frac{\phi_1}{I_2} = \frac{B_2 A_1}{I_2}$ where $B_2 = \frac{\mu_0 I_2 R_2^2}{2(R_2^2 + x^2)^{3/2}}$; then $\phi_2 = M I_1$

Solution:

By reciprocity, imagine a current I_2 flowing in the *bigger* loop and find the (nearly uniform) field it produces at the centre of the small loop:

$$R_2^2 + x^2 = (0.2)^2 + (0.15)^2 = 0.04 + 0.0225 = 0.0625 \Rightarrow (R_2^2 + x^2)^{3/2} = (0.25)^3 = 0.015625$$

$$\frac{B_2}{I_2} = \frac{\mu_0 R_2^2}{2(R_2^2 + x^2)^{3/2}} = \frac{(4\pi \times 10^{-7})(0.04)}{2 \times 0.015625} = \frac{5.027 \times 10^{-8}}{0.03125} = 1.608 \times 10^{-6} \text{ T/A}$$

Area of the small loop:

$$A_1 = \pi r_1^2 = \pi(3 \times 10^{-3})^2 = 2.827 \times 10^{-5} \text{ m}^2$$

$$M = \frac{B_2}{I_2} \times A_1 = (1.608 \times 10^{-6}) \times (2.827 \times 10^{-5}) = 4.546 \times 10^{-11} \text{ H}$$

The actual flux linked with the bigger loop, due to current I_1 in the small loop:

$$\phi_2 = M I_1 = (4.546 \times 10^{-11}) \times 2.0 = 9.1 \times 10^{-11} \text{ Wb}$$

Final Answer

$$\phi_2 = 9.1 \times 10^{-11} \text{ Wb}$$

Common Mistake: Do not try to compute the field of the tiny loop at the far-away centre of the big loop directly – use the reciprocity theorem ($M_{12} = M_{21}$) and compute the (uniform) field of the big loop over the small loop's area instead; it is far simpler and avoids errors.